

STAT

This is where you will go to enter your data.

- 1: Edit – Access to your lists
- 2: SortA(– Sorts your list in ascending order (low to high)
- 3: SortD(– Sorts your list in descending order (high to low)
- 4: ClearList – Clears your list

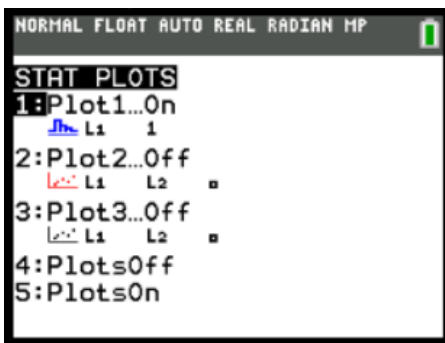
A screenshot of a TI-84 calculator screen showing the list editor for L1. The top status bar shows 'NORMAL FLOAT AUTO REAL RADIAN HP' and a battery icon. The list editor has columns for L1, L2, L3, L4, L5, and a cursor column. The L1 column contains the following values: 47, 48, 49, 50, 56, 59, 59, 65, 72, 79. The cursor is in the first row of the L1 column. Below the list editor, it says 'L1(33)='.

L1	L2	L3	L4	L5	
47					
48					
49					
50					
56					
59					
59					
65					
72					
79					

Enter your data into your list by having **1: Edit** highlighted and clicking [enter]

You can enter multiple lists if you need to. Take your time entering the data!

2nd → STAT PLOT



This is where you will go to graph your data as a histogram or a boxplot.

You can have 3 statistic plots on at once, but you usually just want one at a time.

Notice you can turn them all on or all off at the bottom

Types of Graphs

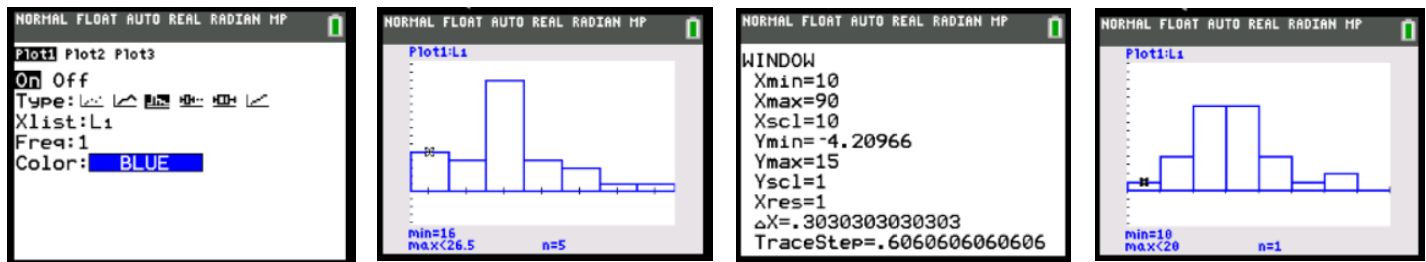
Histogram

Step 1: Have your data entered into your list and highlight the histogram option

Step 2: Go to "ZOOM" then down to 9:ZoomStat

Step 3: The histogram will appear, automatically scaled. Below the graph, the min is the start of the bin and the max is the value the bin goes up to but does not include. The "n" is how many values fall in that interval.

Step 4: You can change your bin width by going to "WINDOW" and adjust the Xmin, Xmax, and Xscl (bin width).

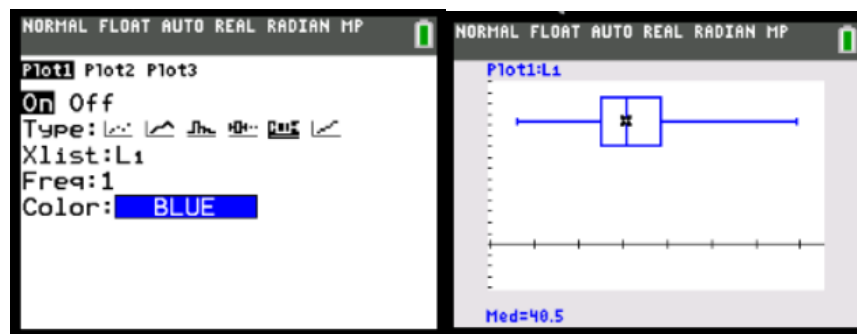


Boxplot

Step 1: Have your data entered into your list and highlight the boxplot option

Step 2: Go to "ZOOM" then down to 9:ZoomStat

Step 3: Your boxplot will appear, automatically fitted to the screen. When you click "Trace", the cursor will rotate through your 5 number summary.



Modified Boxplot

Step 1: Have your data entered into your list and highlight the modified boxplot option.

Step 2: Go to "ZOOM" then down to 9:ZoomStat

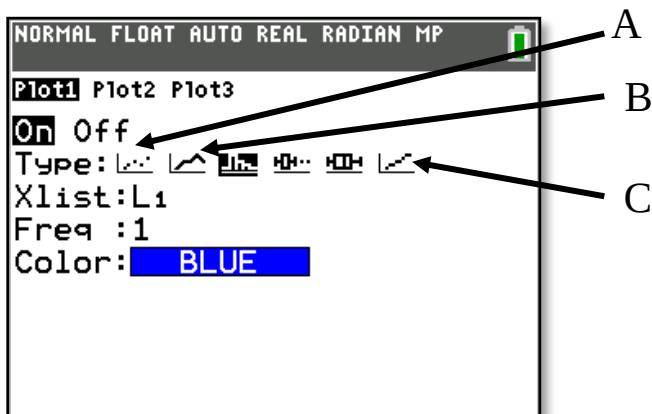
Step 3: You will see your boxplot, with the outliers (calculated by the IQR rule) as dots. If you have no outliers in your data set, it will look like a "regular" boxplot.

Other Graphs

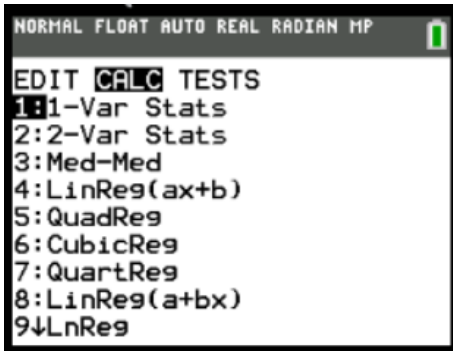
A: Scatterplot – we will cover this option in Unit 2

B: Line Graph – This is for two variable data, but we do not cover it in AP Statistics

C: Normal Probability Plot – This is no longer covered in AP Statistics; it can be used to see if data “fits” a normal model



Summary Statistics

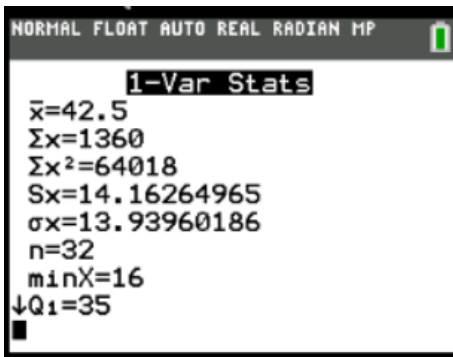
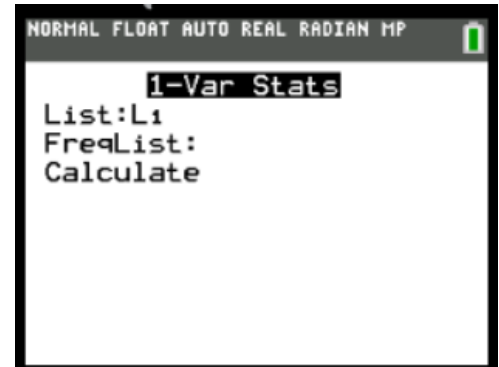


Go to STAT → CALC

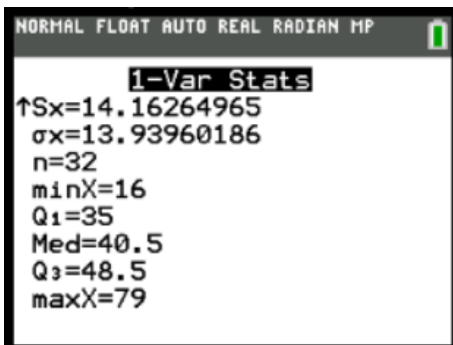
1: 1-Var Stats stands for “1 variable statistics”

Your list will be where your data is located and for FreqList, you can leave it blank. This would be used if you had a list of values and then a list for how often it took those values.

Press Calculate



- $\bar{x}$ is the sample mean
- Σx is the sum of all the values in the list
- Σx^2 is the sum of the values in the list after each of them was squared
- Sx is the sample standard deviation
- σx is the population standard deviation (you will rarely use this! Ignore for now!!)
- n is the total number of items in your list



The last 5 numbers (after you scroll down) are the 5 number summary.

The Normal Distribution



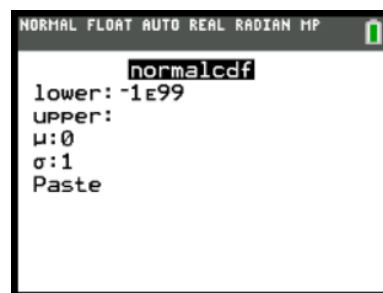
2nd → VARS

DISTR stands for Distribution and it is where you go for your Normal Calculations.

1:normalpdf(→ DO NOT USE THIS. The AP exam does not cover probability density functions. This function returns the probability of a single value of the random variable x , which is the y -coordinates of the normal curve. You would cover this in a second statistics college course.

2: normalcdf(→ this will calculate the proportion (area, probability, etc) of values that fall between the lower and upper boundaries.

- infinity/no boundary is 1e99 or -1e99
- Z-Scores:
lower/upper: z-scores
 $\mu = 0$ and $\sigma = 1$
- Data:
lower/upper: data
 μ = mean of population and σ = standard deviation of population



Ex. The distribution of heights of young women aged 18 to 24 ~ $N(64.5, 2.5)$

$P(x < 66)$

Data:	Z-scores:
lower: -1e99	lower: -1e99
upper: 66	upper: $(66 - 64.5) / 2.5$
μ : 64.5	μ : 0
σ : 2.5	σ : 1

Note: For Unit 1, think of the " $P()$ " notation as "percent" or "proportion". Starting in Unit 4, we will talk about how it is actually "probability" and we will tie these calculations in with proportions 😊

$P(x > 65.3)$

Data:	Z-scores:
lower: 65.3	lower: $(65.3 - 64.5) / 2.5$
upper: 1e99	upper: 1e99
μ : 64.5	μ : 0
σ : 2.5	σ : 1

$P(62.9 < x < 65.9)$

Data:	Z-scores:
lower: 62.9	lower: $(62.9 - 64.5) / 2.5$
upper: 65.9	upper: $(65.9 - 64.5) / 2.5$
μ : 64.5	μ : 0
σ : 2.5	σ : 1

3: invNorm(→ this will calculate the z-score or data value of the distribution that a certain area/percent fall below (ONLY TO LEFT) *Unless you have a fancy calculator

Ex. The distribution of heights of young women aged 18 to 24 ~ N(64.5, 2.5)

What is the 75th percentile?

<u>Data:</u>	<u>Z-scores:</u>
area: .75	area: .75
μ : 64.5	μ : 0
σ : 2.5	σ : 1
>Gives x-value	>Gives z-score

